

Group Project Implementation Summary

Complete Materials Package - Ready to Deploy

Date: November 8, 2025 **Status:** All materials complete and tested

What Was Created

1. Group Project Overview PDF (2 pages)

File: docs/group_project_overview.pdf

Purpose: Main reference document posted on Canvas

Contents: - Project overview (AI comparison focus) - Group assignments (4 groups, datasets assigned) - Complete timeline (Weeks 7-10 + Finals Week) - Grading breakdown (Report 15% + Presentation 10% = 25%) - Deliverable requirements - Resources & tips

Page count: Exactly 2 pages

Distribution: Post to Canvas Sunday, students reference throughout Weeks 7-10

2. Week 7 Assignment

File: docs/week7_assignment.md

Structure: - **Part A:** 5 MC questions (5 pts) - Easy questions on nonparametric methods - **Part B:** Team Plan check-in (5 pts) - Group leader submits

Team Plan Requirements: 1. Team roster with contact info 2. Communication plan 3. Meeting schedule 4. Division of responsibilities 5. Preliminary timeline

Grading: Completion-based (5 pts for reasonable plan)

Due: Sunday Week 7, 11:59pm

3. Week 8 Assignment

File: docs/week8_assignment.md

Structure: - **Part A:** 9 MC questions (9 pts) - Easy questions on contingency tables - **Part B:** Progress Update (1 pt) - Group leader submits

Progress Update: - 3 yes/no questions (met? loaded data? consulted AI?) - Optional text box for blockers

Special Feature: Office Hours Day reminder (Wed 12-1:20pm)

Grading: 1 pt for completion

Due: Sunday Week 8, 11:59pm

4. Week 9 Assignment

File: docs/week9_assignment.md

Structure: - **Part A:** 7 MC questions (7 pts) - Easy questions on correlation/regression - **Part B:** Presentation Readiness (3 pts) - Group leader submits

Presentation Readiness Options: - Option A: Presentation outline - Option B: Draft slides - Option C: Presentation plan

Grading: 3 pts for demonstrating preparation

Due: Sunday Week 9, 11:59pm

Final Dataset Lineup (Optimized for Divergence)

Group	Dataset	N	Divergence	Students	Key Feature
1	Anorexia Treatment	72	8/10	1,4,10,12,13	Pre-post design
2	ToothGrowth	60	8/10	7,11,14,15,16	Factorial 2×3
3	Bacteria	220	9/10	3,5,8,18	Repeated measures!
4	Pima Diabetes	200	7/10	2,6,9,17	Binary outcome

Average Divergence: 8.0/10 **Very High Divergence (8+):** 75% of groups

Grading Structure (Final)

Course Grading Breakdown

Component	Weight	Details
Weekly Assignments	60%	Includes Week 7,8,9 check-ins
Group Project	25%	Report (15%) + Presentation (10%)
Class Participation	15%	
TOTAL	100%	

Weekly Assignments (Weeks 7,8,9)

- **Week 7:** MC (5 pts) + Team Plan (5 pts) = 10 pts total
- **Week 8:** MC (9 pts) + Progress Update (1 pt) = 10 pts total
- **Week 9:** MC (7 pts) + Presentation Readiness (3 pts) = 10 pts total

Check-ins count toward 60% weekly assignment grade, NOT the 25% project grade.

Group Project (25%)

Final Report (15% of course grade): - 5-8 pages + R code appendix - Due: Finals Week Wednesday 11:59pm - Sections: Intro, Methods, Results, **AI Comparison** (1.5-2 pages), Discussion, Appendix

Presentation (10% of course grade): - 15 minutes + 5 min Q&A - Monday Week 10: Groups 1 & 2 - Wednesday Week 10: Groups 3 & 4 - All members must participate

Complete Timeline

Week	Monday	Wednesday	Assignment Due Sunday
Week 7	Regular lecture	Regular lecture	MC (5) + Team Plan (5)
Week 8	Regular lecture	OFFICE HOURS	MC (9) + Progress (1)
		DAY	
Week 9	Regular lecture	Regular lecture	MC (7) + Presentation (3)
Week 10	Presentations:	Presentations:	-
	Groups 1,2	Groups 3,4	
Finals	-	Final Report Due	-
Week		11:59pm	

File Structure

Group_project_John/

docs/

```

group_project_overview.tex    # 2-page LaTeX source
group_project_overview.pdf    # ← POST TO CANVAS
week7_assignment.md           # Week 7 Canvas assignment text
week8_assignment.md           # Week 8 Canvas assignment text
week9_assignment.md           # Week 9 Canvas assignment text

```

group_1/

```

anorexia.csv
divergence_test_anorexia.R (instructor reference)
group_1_roster.csv

```

group_2/

```

toothgrowth.csv
divergence_test_toothgrowth.R (instructor reference)
group_2_roster.csv

```

group_3/

```

bacteria.csv    NEW - 9/10 divergence!
divergence_test_bacteria.R (instructor reference)
group_3_roster.csv

```

group_4/

```

pima_diabetes.csv
divergence_test_pima.R (instructor reference)
group_4_roster.csv

```

```

group_assignments.csv    # Student → group mapping
random_group_assignment.R # Script used for randomization
DIVERGENCE_SUMMARY.md    # Complete divergence analysis
IMPLEMENTATION_SUMMARY.md # This file

```

Deployment Instructions

Sunday (This Week) - Project Launch

1. Upload to Canvas Files:

- `group_project_overview.pdf` → Post prominently
- Create folder: `Group_Project_Data/`
 - Upload `group_1/anorexia.csv`
 - Upload `group_2/toothgrowth.csv`
 - Upload `group_3/bacteria.csv`
 - Upload `group_4/pima_diabetes.csv`
- Upload data dictionary (if separate file)

2. Create Canvas Announcement:

GROUP PROJECT BEGINS - READ NOW!

The H524 Group Project starts this week! You'll analyze medical data while comparing how Claude, ChatGPT, and Microsoft Copilot approach the same statistical questions.

READ FIRST: `Group_Project_Overview.pdf` [link]

Your Groups (randomly assigned):

- Group 1: Students 1, 4, 10, 12, 13 - Anorexia Treatment
- Group 2: Students 7, 11, 14, 15, 16 - ToothGrowth
- Group 3: Students 3, 5, 8, 18 - Bacteria Treatment
- Group 4: Students 2, 6, 9, 17 - Pima Diabetes

Week 7 Assignment (due Sunday) includes your first check-in: Group leader submits a team plan.

Questions? Post to Canvas Discussions.

3. Assign Students to Canvas Groups:

- Use Canvas Groups feature
- Create 4 groups
- Assign students per `group_assignments.csv`
- Designate one student in each group as “leader” (alphabetically first?)

Week 7 - Create Canvas Assignment

Title: Week 7 Assignment

Instructions: Copy text from `week7_assignment.md`

Points: 10 (5 MC + 5 Check-in)

Submission Type: - Part A: Quiz (MC questions) - Part B: Text entry OR file upload

Due: Sunday 11:59pm

Week 8 - Create Canvas Assignment

Title: Week 8 Assignment

Instructions: Copy text from `week8_assignment.md`

Points: 10 (9 MC + 1 Check-in)

Submission Type: - Part A: Quiz (MC questions) - Part B: Text entry (brief)

Due: Sunday 11:59pm

Also: Create Canvas announcement for Office Hours Day (Wed 12-1:20pm)

Week 9 - Create Canvas Assignment

Title: Week 9 Assignment

Instructions: Copy text from `week9_assignment.md`

Points: 10 (7 MC + 3 Check-in)

Submission Type: - Part A: Quiz (MC questions) - Part B: File upload OR text entry

Due: Sunday 11:59pm

Week 10 - Presentations

Monday: Groups 1 & 2 present (during class 12-1:20pm)

Wednesday: Groups 3 & 4 present (during class 12-1:20pm)

Grading: Use presentation rubric (create separate or embedded in gradebook)

Finals Week

Final Report Due: Wednesday 11:59pm

Submission: Canvas assignment (file upload, PDF)

Grading: Use report rubric (separate document recommended)

Quality Assurance Completed

Datasets: - All 4 datasets exported to CSV - Divergence testing completed (actual R results documented)
- Multiple valid approaches verified for each - Average divergence 8.0/10 (target: 8+/10)

Materials: - Project overview is exactly 2 pages - Week 7,8,9 assignments include ONLY MC + check-ins
- Grading structure sums to 100% - Timeline is complete and realistic - Check-ins are part of weekly assignments (60%), not project (25%)

Student Experience: - Clear instructions at each stage - Scaffolded support (3 check-ins) - Office Hours Day Week 8 - Emphasis on AI comparison (not just analysis)

Expected Student Workload

Week 7: - Weekly assignment: 30 min (easy MC + team formation) - Initial project work: 1-2 hours (meet, download data, plan) - **Total: 1.5-2.5 hours**

Week 8: - Weekly assignment: 20 min (easy MC + quick check-in) - Project work: 4-6 hours (data analysis, AI consultation) - **Total: 4.5-6.5 hours**

Week 9: - Weekly assignment: 30 min (easy MC + outline submission) - Project work: 4-6 hours (AI comparison, presentation prep) - **Total: 4.5-6.5 hours**

Week 10: - Presentation: 1-2 hours (final prep + delivery) - **Total: 1-2 hours**

Finals Week: - Final report: 3-5 hours (writing, formatting, polishing) - **Total: 3-5 hours**

Grand Total: 14-22 hours (appropriate for 25% of course grade over 5 weeks)

Key Pedagogical Features

1. **AI Comparison is Central** (not peripheral)
 - 30-40% of presentation time
 - 1.5-2 pages of report (out of 5-8)
 - Explicit grading weight
 2. **Scaffolded Support**
 - Week 7: Team formation
 - Week 8: Progress check + Office Hours Day
 - Week 9: Presentation readiness
 - Not high-stakes until Week 10
 3. **Realistic Timeline**
 - 5 weeks start to finish
 - Check-ins keep groups accountable
 - Finals Week for final report (after presentations)
 4. **High Divergence Datasets**
 - 75% of groups have Very High divergence (8+/10)
 - Bacteria dataset (9/10) tests critical independence assumption
 - All datasets validated with actual R testing
 5. **Flexible Grading**
 - Check-ins completion-based (low stakes)
 - Report and presentation are main deliverables (25% course grade)
 - Clear rubric potential for objective grading
-

Instructor Notes

Grading Load: - Week 7: 4 team plans to review (~20 min total) - Week 8: 4 brief updates to scan (~5 min total) - Week 9: 4 outlines to review (~15 min total) - Week 10: 4 presentations to attend/grade (~90 min total) - Finals Week: 4 reports to grade (~4-6 hours total)

Total instructor time: ~6-7 hours over 5 weeks

Office Hours Day (Week 8 Wed): - Be available 12:00-1:20pm - Classroom first, then office - Expect 1-3 groups to attend (not all)

Common Questions to Prepare For: - “Which AI tool is best?” → There’s no single answer, that’s the point! - “My AI suggested method X, is that right?” → Check assumptions, verify with R - “What if all three AI tools disagree?” → Perfect! Document differences, evaluate critically

Project Success Metrics

For Students: - Complete all 3 check-ins (demonstrated progress) - Submit final report on time - Deliver presentation with all members participating - Document meaningful AI comparison

For Instructor: - 100% of groups submit final deliverables - Presentations show evidence of AI comparison (not just analysis) - Reports demonstrate critical evaluation (not blind acceptance of AI) - Students learn AI verification strategies

STATUS: READY FOR DEPLOYMENT

All materials complete and validated. Ready to post Sunday!